

Inverse problems for stochastic partial differential equations

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1. Introduction

Formulation of inverse problems for SPDEs

Let $T > 0$, $G \subset \mathbb{R}^n$ ($n \in \mathbb{N}$) be a given bounded domain with a C^2 boundary Γ . Denote $Q = (0, T) \times G$ and $\Sigma = (0, T) \times \Gamma$.

Consider the following stochastic hyperbolic equation:

$$\begin{cases} dy_t - \Delta y dt = (ay + f)dt + (by + g)dW(t) & \text{in } Q, \\ y = \varphi & \text{on } \Sigma, \\ y(0) = y_0, \quad y_t(0) = y_1 & \text{in } G, \end{cases} \quad (1)$$

where a , b , f , g and φ are suitable functions/stochastic processes and $W(t)$ denotes the standard Brownian motion.

Direct problem: Determine the solution y of system (1) by specifying $(a, b, f, g, \varphi, y_0, y_1)$ along with Q . This can be achieved using classical well-posedness results of stochastic evolution equations.

Inverse problem: Involves certain unknown components in $(a, b, f, g, \varphi, y_0, y_1)$ and Q , which need to be determined through measuring some information about the solution y :

$$\mathcal{M}(y)(t, x) = h(t, x), \quad (t, x) \in \mathcal{O} \subset [0, T] \times \overline{G}.$$

Here, $\mathcal{M}(\cdot)$ is a known operator, representing the method of obtaining measurement data, and $\mathcal{O}$ denotes the position of the measurement.

Three Key Concerns:

- Can measurements **uniquely determine** the unknowns?
- Is **stable reconstruction** feasible?
- Are there **efficient numerical algorithms**?

Deterministic Case: Uniqueness Fails

Consider the following inverse source problem:

$$\begin{cases} y_{tt} - \Delta y = f & \text{in } Q, \\ y = 0 & \text{on } \Sigma, \\ y(0) = y_t(0) = 0 & \text{in } G. \end{cases} \quad (2)$$

Inverse Problem: Can we uniquely determine f from the measurements $\frac{\partial y}{\partial \nu}|_{\Sigma}$ and $y(T)$?

Answer: NO.

For any $y^* \in C_0^\infty(Q)$, let $f = y_{tt}^* - \Delta y^*$. Then y^* solves (2) with $\frac{\partial y^*}{\partial \nu}|_{\Sigma} = 0$, $y^*(T) = 0$. But we can choose y^* such that $f \neq 0$ in Q .

Stochastic Case: Uniqueness Holds

Consider the analogous problem in the stochastic setting:

$$\begin{cases} dz_t - \Delta z dt = g dW(t) & \text{in } Q, \\ z = 0 & \text{on } \Sigma, \\ z(0) = z_t(0) = 0 & \text{in } G. \end{cases}$$

Inverse Problem: Can we uniquely determine g from the measurements $\frac{\partial z}{\partial \nu}|_{\Sigma}$ and $z(T)$?

Answer: YES. One can show that

$$\frac{\partial z}{\partial \nu}|_{\Sigma} = 0 \quad \text{and} \quad z(T) = 0 \quad \implies \quad g = 0 \quad \text{in } Q, \quad \mathbb{P}\text{-a.s.}$$

(Lü and Zhang, *Comm. Pure Appl. Math.*, 2015)

2. Inverse State Problem for Stochastic Hyperbolic Equations

Joint work with Qi Lü

Semilinear Stochastic Hyperbolic Equation

Let $T > 0$, and let $(\Omega, \mathcal{F}, \mathbf{F}, \mathbb{P})$ be a complete filtered probability space, where $\mathbf{F} = \{\mathcal{F}_t\}_{t \geq 0}$ is the natural filtration generated by a one-dimensional standard Brownian motion $\{W(t)\}_{t \geq 0}$.

Consider the following semilinear stochastic hyperbolic equation:

$$\begin{cases} du_t - \sum_{j,k=1}^n (b^{jk} u_{x_j})_{x_k} dt = F(u, u_t, \nabla u) dt + a u dW(t) & \text{in } Q, \\ u_t(0) = 0 & \text{in } G, \\ u(0) = u_0 & \text{in } G, \end{cases} \quad (3)$$

Coefficients: Let $(b^{jk})_{1 \leq j, k \leq n} \in C^1(G; \mathbb{R}^{n \times n})$ satisfy that $b^{jk} = b^{kj}$ for all $j, k = 1, \dots, n$ and for some constant $s_0 > 0$,

$$\sum_{j,k=1}^n b^{jk} \xi_j \xi_k \geq s_0 |\xi|^2, \text{ for all } (x, \xi) \in G \times \mathbb{R}^n.$$

Furthermore, $a \in L^\infty_{\mathbb{F}}(0, T; L^\infty(G))$.

Nonlinearity: Let $F : [0, T] \times \Omega \times G \times \mathbb{R} \times \mathbb{R} \times \mathbb{R}^n \rightarrow \mathbb{R}$ be such that, for each $(\eta, \rho, \zeta) \in \mathbb{R} \times \mathbb{R} \times \mathbb{R}^n$, $F(\cdot, \cdot, \cdot, \eta, \rho, \zeta) : [0, T] \times \Omega \times G$ is an $L^2(G)$ -valued $\mathbf{F}$ -adapted process, and there exists a constant L such that for a.e. $(t, \omega, x) \in [0, T] \times \Omega \times G$ and any $(\eta_i, \varrho_i, \zeta_i) \in \mathbb{R} \times \mathbb{R} \times \mathbb{R}^n$ ($i = 1, 2$),

$$|F(t, \omega, x, \eta_1, \varrho_1, \zeta_1) - F(t, \omega, x, \eta_2, \varrho_2, \zeta_2)| \leq L(|\eta_1 - \eta_2| + |\varrho_1 - \varrho_2| + |\zeta_1 - \zeta_2|_{\mathbb{R}^n});$$

and $|F(\cdot, \cdot, \cdot, 0, 0, 0)| \in L^2_{\mathbb{F}}(0, T; L^2(G))$.

Inverse State Problem

Problem

Determine the initial state u_0 from the lateral Cauchy data $u|_{\Sigma}$ and $\frac{\partial u}{\partial \nu}|_{\Sigma_0}$.

- $\Sigma = (0, T) \times \Gamma$, $\Sigma_0 = (0, T) \times \Gamma_0$ where $\Gamma_0 \subset \Gamma$ is a subset of the boundary.
- Only partial boundary data is available.

Conditions 1. There exists a positive function $\psi(\cdot) \in C^2(\overline{G})$ satisfying the following:

- For some constant $\mu_0 > 0$ and any $(x, \xi) \in \overline{G} \times \mathbb{R}^n$, it holds that

$$\sum_{j,k=1}^n \left\{ \sum_{j',k'=1}^n \left[2b^{jk'} (b^{j'k} \psi_{x_{j'}})_{x_{k'}} - b_{x_{k'}}^{jk} b^{j'k'} \psi_{x_{j'}} \right] \right\} \xi_j \xi_k \geq \mu_0 \sum_{j,k=1}^n b^{jk} \xi_j \xi_k.$$

- There is no critical point of $\psi(\cdot)$ in $\overline{G}$, i.e., $\min_{x \in \overline{G}} |\nabla \psi(x)| > 0$.

Put

$$\Gamma_0 \triangleq \left\{ x \in \Gamma \mid \sum_{j,k=1}^n b^{jk}(x) \psi_{x_j}(x) \nu^k(x) > 0 \right\}.$$

Theorem (Lü, Inverse Problems, 2013)

Under Conditions 1, there exists $T_0 > 0$ such that for $T > T_0$, there exists constant $C > 0$ so that for all $u_0, \hat{u}_0 \in H_0^1(G)$, the solutions satisfy

$$|u_0 - \hat{u}_0|_{H_0^1(G)} \leq C \left| \frac{\partial u}{\partial \nu} - \frac{\partial \hat{u}}{\partial \nu} \right|_{L_{\mathbb{F}}^2(0, T; L^2(\Gamma_0))}.$$

- **Identifiability:** The initial state u_0 is uniquely determined by the boundary data.
- **Stability:** The initial state depends continuously on the boundary data.
- **Reconstruction:** Is it possible to reconstruct the initial state from the boundary data?

Iterative Regularization: Fixed-Point Scheme

Idea: Construct a sequence $\{u_n\}$ by minimizing the weighted Tikhonov functional at each iteration.¹

- **Initialization:** Start from any u_0 in the admissible set.
- **Iteration:** For $n \geq 0$, set

$$u_{n+1} = \mathfrak{G}(u_n) \stackrel{\Delta}{=} \arg \min_{\varphi \in \mathcal{U}} J(\varphi; u_n).$$

- No need for a good initial guess

¹Nguyen and Klibanov, *Inverse Problems*, 2022.

Carleman Estimate

Carleman estimates are weighted energy inequalities which can be used for proving uniqueness, stability, and convergence in inverse problems.

Intuitive example: Let $\tau > 0$. It holds that

$$2\tau \int_{-1}^1 |u|^2 e^{\tau t^2} dt \leq \int_{-1}^1 \left| \frac{du}{dt} \right|^2 e^{\tau t^2} dt, \quad \forall u \in C_0^\infty(-1, 1) \quad (4)$$

Proof. Set $v(t) = u(t)e^{\tau t^2/2}$. Then we have

$$\begin{aligned} |u'(t)|^2 e^{\tau t^2} &= |v'(t) - \tau t v(t)|^2 = v'(t)^2 - 2\tau t v(t)v'(t) + \tau^2 t^2 v(t)^2 \\ &= |v'(t) + \tau t v(t)|^2 - 4\tau t v(t)v'(t) \\ &= |v'(t) + \tau t v(t)|^2 + 2\tau |v(t)|^2 - 2\tau (tv(t)^2)' \end{aligned}$$

Integrate this over $[-1, 1]$, noting that $v \in C_0^\infty(-1, 1)$, we get (4).

Carleman Weight Function

- **Weight function:** For $c_0, \lambda > 0$,

$$\theta = e^\ell, \quad \ell(t, x) = \lambda(\psi(x) - c_1 t^2)$$

- **Boundary splitting:** Γ_0 defined by

$$\Gamma_0 \triangleq \left\{ x \in \Gamma \mid \sum_{j,k=1}^n b^{jk}(x) \psi_{x_j}(x) \nu^k(x) > 0 \right\}.$$

- For $du_t - \Delta u dt$, function $\psi(x) = |x - x_0|^2$ with $x_0 \in \mathbb{R}^n \setminus \overline{G}$ satisfies the conditions and $\Gamma_0 = \{x \in \Gamma \mid (x - x_0) \cdot \nu > 0\}$.

Statement of the Carleman Estimate

Theorem (Lü and Wang, Inverse Problems, 2025)

Under Conditions 1, there exist positive constants C and λ_0 such that for all $\lambda \geq \lambda_0$, the solution z to the stochastic hyperbolic equation:

$$dz_t - \sum_{j,k=1}^n (b^{jk} z_{x_j})_{x_k} dt = Ydt + azdW(t)$$

with $z_t(0) = 0$, $z = 0$ on Σ , $\frac{\partial z}{\partial \nu} = 0$ on Σ_0 satisfies:

$$\mathbb{E} \int_Q \theta^2 (\lambda z_t^2 + \lambda |\nabla z|^2 + \lambda^3 z^2) dx dt \leq C \mathbb{E} \int_G \theta^2 (\lambda |\nabla z|^2 + \lambda^3 |z|^2)_{t=T} dx + C \mathbb{E} \int_Q \theta^2 |Y|^2 dx dt.$$

Tikhonov Regularization Functional

- At each step, consider

$$J_n(\varphi) = |\mathcal{P}\varphi - \mathbf{IF}(u_n)|_{L_{\mathbb{F}}^{2,w}(\Omega; H^1(0,T; L^2(G)))}^2 + \kappa |\varphi|_{\mathcal{H}_{\lambda,0}^2}^2.$$

- $\mathcal{P}\varphi = \varphi_t(t, x) - \int_0^t \sum_{j,k=1}^n (b^{jk}(s, x) \varphi_{x_j}(s, x))_{x_k} ds - \int_0^t a(s, x) \varphi(s, x) dW(s),$
 $\mathbf{IF}(u) = \int_0^t F(u, u_t, \nabla u) ds.$
- Admissible set:

$$\mathcal{U} \triangleq \left\{ \varphi \in \mathcal{H}^2 \mid \mathcal{P}\varphi \in L_{\mathbb{F}}^2(\Omega; H^1(0, T; L^2(G))), \quad \varphi_t(0) = 0, \quad \varphi|_{\Sigma} = f, \quad \frac{\partial \varphi}{\partial \nu} \Big|_{\Sigma_0} = g \right\},$$

$$L_{\mathbb{F}}^{2,w}(0, T; H^p(G)) = \{ \varphi \in L_{\mathbb{F}}^2(0, T; H^p(G)) \mid \mathbb{E} \int_0^T \int_G \theta^2 |D^\alpha \varphi|^2 dx dt < \infty, \quad |\alpha| \leq p \},$$

$$\mathcal{H}_{\lambda, c_0}^p = L_{\mathbb{F}}^{2,w}(0, T; H^p(G)) \cap L_{\mathbb{F}}^{2,w}(\Omega; H^1(0, T; H^1(G))).$$

Global Convergence result

Theorem (Lü and Wang, Inverse Problems, 2025)

Under Conditions 1, fix $\gamma \in (0, 1)$ and $\kappa > 0$. There exist positive constants C and $\tilde{\lambda}_0(\kappa)$ such that for all $\lambda \geq \tilde{\lambda}_0$, $\delta < e^{-C\lambda/\gamma}$, and $n \in \mathbb{N}$, the sequence $\{u_n\}$ satisfies

$$|u_{n+1} - u^*|_{\mathcal{H}_{\lambda, c_0}^1}^2 \leq \left(\frac{C}{\lambda}\right)^{n+1} |u_0 - u^*|_{\mathcal{H}_{\lambda, c_0}^1}^2 + C(\delta^{2-2\gamma} + \kappa |u^*|_{\mathcal{H}_{\lambda, 0}^2}^2),$$

where u^* is the true solution.

- Convergence is geometric in n .
- $\delta > 0$ quantifies the noise level in the measurements: for $\mathcal{E} \in \mathcal{H}^2$,

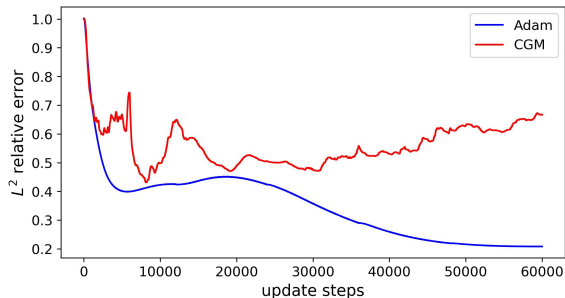
$$|\mathcal{P}\mathcal{E}|_{L_{\mathbb{F}}^2(0, T; H^1(0, T; L^2(G)))} + |\mathcal{E}|_{\mathcal{H}^2} \leq \delta, \quad f|_{\Sigma} = f^* + \mathcal{E}|_{\Sigma}, \quad g = g^* + \frac{\partial \mathcal{E}}{\partial \nu} \Big|_{\Sigma_0}.$$

Numerical Algorithm: Discretization and Optimization

- **Discretization:** The spatial and temporal domains are discretized using finite difference schemes, with the Euler-Maruyama method applied for stochastic integration.
- **Functional:** The weighted Tikhonov functional is discretized in accordance with the chosen grid.
- **Optimization:** At each iteration, the Adam (Adaptive Moment Estimation) optimizer is employed to minimize the discretized functional.
- **Gradient computation:** Gradients are efficiently and accurately computed via automatic differentiation.

Why Adam and Autodiff?

- **Adam** is well-suited for high-dimensional, non-convex, and stochastic optimization tasks, making it effective for minimizing the regularized functional in our setting.
- **Automatic differentiation** enables efficient and accurate gradient computation, eliminating the need to explicitly solve backward stochastic partial differential equations.



Numerical example

- Consider the following semilinear stochastic hyperbolic equation:

$$\begin{cases} du_t - \Delta u dt = F(u, u_t, \nabla u) dt + a u dW(t) & \text{in } Q, \\ u_t(0) = 0 & \text{in } G, \\ u(0) = u_0 & \text{in } G, \end{cases}$$

where $G = (0, 1) \times (0, 1.5)$ and $T = 1$.

- The observation boundary

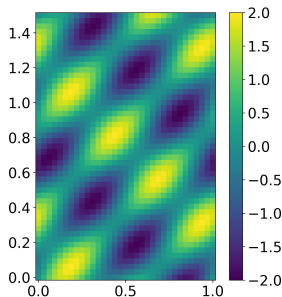
$$\Gamma_0 = \partial G \setminus \{(x, 0) \mid x \in [0, 1]\}$$

- Multiplicative noise

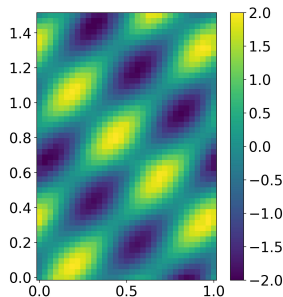
$$f = f^*(1 + \delta\xi), \quad g = g^*(1 + \delta\xi),$$

where ξ is a random variable uniformly distributed in $[-1, 1]$.

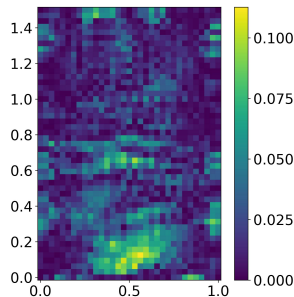
Numerical Example 1



True initial state u_0^*



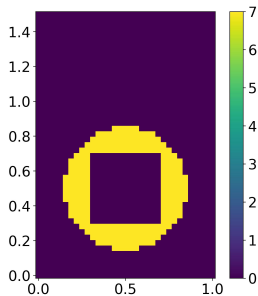
Reconstructed initial state u_0



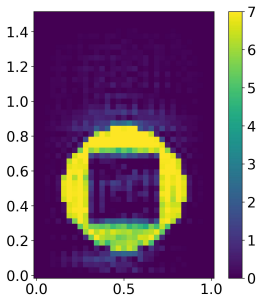
Normalized pointwise error

The final L^2 relative error is 6%. The oscillatory initial state is accurately reconstructed even though $F = \min\{e^u + |\nabla u|, 10\}$ is not globally Lipschitz.

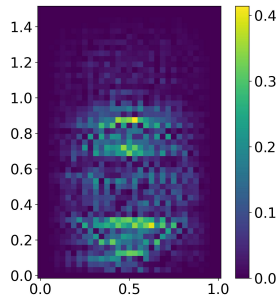
Numerical Example 2



True initial state u_0^*



Reconstructed initial state u_0



Normalized pointwise error

With stochastic boundary data and partial boundary measurements, the final L^2 relative error is 20.7%. The largest discrepancies occur near the initial-state interface and the unobserved boundary.

Takeaways from the hyperbolic problem

- **Accurate and robust:** The method reconstructs initial states reliably under noise, nonlinearity, and partial data.
- **Modern optimization (Adam, autodiff)** enables efficient and robust computation.
- **Carleman-based regularization** is a powerful tool for the inverse problems of SPDEs.

3. Geometric Inverse Source Problem for a Stochastic Heat Equation

Joint work with Yunzhang Li, Qi Lü, and Meizhi Qian

Two geometric source channels

Let $G \subset \mathbb{R}^d$, $d \geq 2$, be a bounded C^2 domain, let $\Gamma_1 \subset \partial G$ be a nonempty relatively open set, and let $K \Subset G$ be compact. Consider

$$\begin{cases} du = (\Delta u + \chi_{D_0} + au) dt + (bu + \chi_{D_1}) dW(t) & \text{in } (0, T) \times G, \\ u = 0 & \text{on } (0, T) \times \partial G, \\ u(0) = u_0 & \text{in } G, \end{cases}$$

where $D_0, D_1 \subset K$ are measurable sets.

Geometric inverse source problem

Determine (D_0, D_1) from the partial boundary flux $\partial u / \partial n$ on $(0, T) \times \Gamma_1$.

Given noisy data g , we minimize

$$\mathcal{J}(D_0, D_1) = \frac{1}{2} \mathbb{E} \int_0^T \int_{\Gamma_1} \left(\frac{\partial u}{\partial n} - g \right)^2 d\sigma dt + \beta_0 \mathcal{P}_G(D_0) + \beta_1 \mathcal{P}_G(D_1).$$

Uniqueness of the two supports

Theorem (Li, Lü, Qian, and Wang, arXiv, 2026)

Assume $a(t, x, \omega) = a_0(t, \omega) + c(x)$ and $b \in L^{\infty}_{\mathbb{F}}(0, T; W^{1, \infty}(G))$. Let u^1, u^2 correspond to (D_0^1, D_1^1) and (D_0^2, D_1^2) , with the same coefficients and initial datum. If

$$\frac{\partial u^1}{\partial n} = \frac{\partial u^2}{\partial n} \quad \text{on } (0, T) \times \Gamma_1,$$

then $\chi_{D_0^1} = \chi_{D_0^2}$ and $\chi_{D_1^1} = \chi_{D_1^2}$ a.e. in G .

Proof sketch: An exponential transformation followed by taking expectations identifies the drift source; the Itô isometry and a deterministic adjoint reachability argument identify the diffusion source.

Existence of a minimizer

Theorem (Li, Lü, Qian, and Wang, arXiv, 2026)

The functional

$$\mathcal{J}(D_0, D_1) = \frac{1}{2} \mathbb{E} \int_0^T \int_{\Gamma_1} \left(\frac{\partial u}{\partial n} - g \right)^2 d\sigma dt + \beta_0 \mathcal{P}_G(D_0) + \beta_1 \mathcal{P}_G(D_1),$$

admits a measurable minimizer over all pairs (D_0, D_1) contained in K .

Since the sources are supported away from ∂G , local hidden boundary regularity gives

$$\left\| \frac{\partial z}{\partial n} \right\|_{L^2_{\mathbb{F}}(0,T;L^2(\partial G))} \leq C_M \left(\|f_0\|_{L^2_{\mathbb{F}}(0,T;H^{-1}(G))} + \|f_1\|_{L^2_{\mathbb{F}}(0,T;H^{-1}(G))} + \|z_0\|_{H^1_0(G)} \right),$$

where z solves $dz = (\Delta z + az + f_0) dt + (bz + f_1) dW(t)$ and $z(0) = z_0$.

Shape sensitivity analysis

The backward stochastic adjoint is

$$\begin{cases} dp = -(\Delta p + ap + bq)dt + q dW(t) & \text{in } (0, T) \times G, \\ p = \partial_n u - g & \text{on } (0, T) \times \Gamma_1, \\ p = 0 & \text{on } (0, T) \times (\partial G \setminus \Gamma_1), \\ p(T) = 0 & \text{in } G. \end{cases}$$

Let $D_0, D_1 \Subset G$ be open sets with C^2 boundaries, and let $V_0, V_1 \in C_c^1(G; \mathbb{R}^n)$. Then

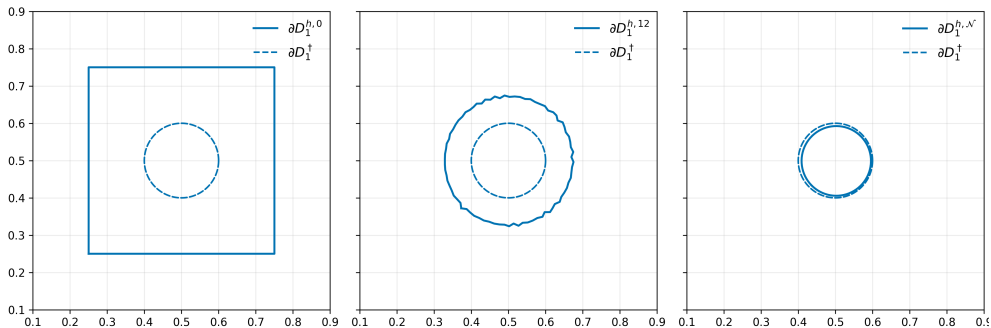
$$\begin{aligned} d\mathcal{J}(D_0, D_1; V_0, V_1) = & -\mathbb{E} \int_0^T \int_{\partial D_0} p V_0 \cdot \nu_0 d\sigma dt + \beta_0 \int_{\partial D_0} \rho_0 V_0 \cdot \nu_0 d\sigma \\ & -\mathbb{E} \int_0^T \int_{\partial D_1} q V_1 \cdot \nu_1 d\sigma dt + \beta_1 \int_{\partial D_1} \rho_1 V_1 \cdot \nu_1 d\sigma. \end{aligned}$$

Here, p and q govern the sensitivities of ∂D_0 and ∂D_1 , respectively.

Shape-gradient reconstruction method

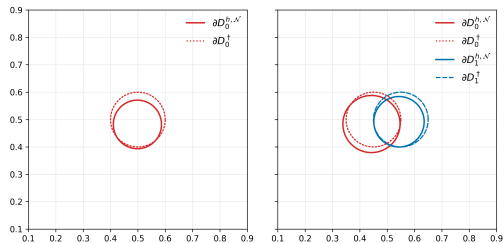
- 1 Solve the stochastic heat equation for the current (D_0, D_1) .
- 2 Form the boundary residual and solve the backward stochastic adjoint.
- 3 Compute the two boundary shape densities from p and q .
- 4 Use an elasticity Riesz map to obtain smooth descent fields.
- 5 Move both polygonal interfaces and regenerate an interface-fitted mesh.

Representative geometric reconstructions I

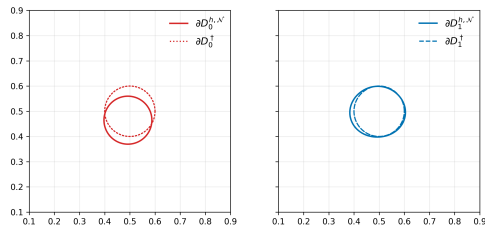


Evolution of the diffusion-support interface at $k = 0$, $k = 12$, and $k = 50$. Dashed curves indicate the target interface; solid curves indicate the reconstructed interface.

Representative geometric reconstructions II



Full-boundary observation.



The bottom side unobserved.

Two complementary inverse-problem frameworks

Inverse state problem

- Function-valued unknown u_0 .
- Carleman estimate plus fixed-point regularization.
- Global convergence and autodiff-based numerics.

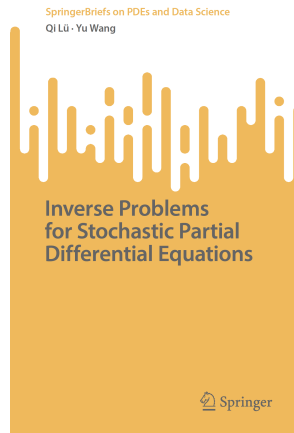
Geometric inverse source problem

- Set-valued unknowns (D_0, D_1) .
- BV compactness, uniqueness, and stochastic shape calculus.
- Two-channel adjoint gradient and LSMC numerics.

Common theme: exploit the Itô stochastic structure rather than treating the problem as a family of deterministic inverse problems.

Selected references

- 1 Q. Lü and Y. Wang, *An inverse source problem for semilinear stochastic hyperbolic equations*, *Inverse Problems* **41** (2025), 115014.
- 2 Y. Li, Q. Lü, M. Qian, and Y. Wang, *A geometric inverse source problem for stochastic parabolic equations*, arXiv:2608.01351, 2026.
- 3 Q. Lü and Y. Wang, *Inverse Problems for Stochastic Partial Differential Equations*, Springer, 2026.



Thank You!